

BIOLOGICAL SEQUENCE MOTIF DISCOVERY

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2024CSM023

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Methods Of Biological Sequence Motifs Discovery

1. Probabilistic Approach
2. Markov Models
3. Exact Methods
4. Suffix Trees And Box Links
5. Random And Uniform Projections

Probabilistic Approach

- There Are Two Various Methods Of Probabilistic Approach:
- **Gibbs Sampling** :- It Involves Calculating a Combined Probability Distribution. This “Probability Distribution” is then used to Generate a Sequence of Samples and Generating Position Weight Matrices (PWM) for these Substrings.
- **Expectation Maximization** :- This Method involves “Random Identification” of some initial PWM, and then Scanning the Set of Input Sequences to Find Matches to the PWM. “MEME ” is a Motif-Discovery Algorithm Based on Expectation Maximization.

Example: A G T C G T C A T G A

Steps:-

Training the Model

For our Example Sequence, we compute the Transition Probabilities as follows:

From A $\rightarrow$ G (1 time)

From G $\rightarrow$ T (1 time)

From T $\rightarrow$ C (2 times)

From C $\rightarrow$ G (1 time)

From G $\rightarrow$ T (1 time)

From T $\rightarrow$ C (1 time)

From C $\rightarrow$ A (1 time)

From A $\rightarrow$ T (1 time)

From T $\rightarrow$ G (1 time)

Transition Matrix

Based on these Observations, we can create a Transition Matrix (ignoring the start/end states for simplicity):

From → To	A	C	G	T
A	0	0	1	1
C	1	0	1	0
G	0	0	0	2
T	0	2	0	0

Prediction

To predict the next symbol in the sequence.

Let's Say the Current Symbol is "C"

From the matrix row for C;

So, the Most Likely Next Symbol After "C" is A or G.

- After A $\rightarrow$ G or T (Equal)
- After G $\rightarrow$ T (Always)
- After T $\rightarrow$ C (Always)
- After C $\rightarrow$ A or G (Equal)

Markov Models

- A “Markov Model” is built by analyzing existing data and building a model that can define well the probability of occurrence of a Nucleotide in some position based on the previous Nucleotides in the sequence.
- Markov Models are built around the assumption that the probability of each state depends only on the previous state.
- Variable Length Markov Models :- These involve developing Markov Models with the Assumption that the existence of a Nucleotide at some position does not always depend on fixed number of Previous Positions.

Example: A G T C G T C A T G A

Steps:-

Training Matrix

From → To	A	C	G	T
A	0	0	1	1
C	1	0	1	0
G	0	0	0	2
T	0	2	0	0

Prediction

From → To	A	C	G	T
A	0.0	0.0	0.5	0.5
C	0.5	0.0	0.5	0.0
G	0.0	0.0	0.0	1.0
T	0.0	1.0	0.0	0.0

Exact Methods

- Although “The Motif Discovery problem is NP-Complete”.
- Exact Methods Aim to find the Optimal Solution, such as aligning two sequences using Dynamic Programming (e.g., Needleman-Wunsch algorithm).
- We are aligning the sequence AGTCA with GTCA. Let's define a scoring system:
Match = +1
Mismatch = -1
Gap = -2

Example: A G T C G T C A T G A

We now build a scoring matrix for the alignment:

Steps:-

Matrix Initialization

A Matrix with Rows Representing AGTCA and Columns representing GTCA.

	-	G	T	C	A
-	0	-2	-4	-6	-8
A	-2	-1	-3	-5	-5
G	-4	0	-2	-4	-6
T	-6	-2	1	-1	-3
C	-8	-4	-1	2	0
A	-10	-6	-3	0	3

Fill the Matrix

Use the scoring system to fill the matrix.

Now we fill the empty cells.

Each cell is filled using this formula:

$$\text{score}(i, j) = \max \begin{cases} \text{Top-left} + \text{match or mismatch} \\ \text{Top} + \text{gap penalty} \\ \text{Left} + \text{gap penalty} \end{cases}$$

Cell (1,1): Compare A vs G

- Diagonal (0,0) = 0 $\rightarrow$ mismatch $\rightarrow 0 + (-1) = -1$.
- Top (0,1) = -2 $\rightarrow -2 + (-2) = -4$.
- Left (1,0) = -2 $\rightarrow -2 + (-2) = -4$.

Max = -1.

Backtracking

Once the matrix is filled, we backtrack from the bottom right corner to find the optimal alignment.

Optimal Alignment:

Seq1: AGTCA

Seq2: -GTCA

The final score is 3, and optimal alignment is:-

AGTCA

-GTCA

Suffix-Trees and Box-Links

- Algorithms using Suffix Trees and Box-Links have also been Proposed for the Motif Discovery Problem.
- Algorithms based on “Box-Link” are Similar to the “Suffix Tree”. Here, instead of having a Boolean array associated with each node, a Box-link is associated with it. A Box-link is a Data Structure that contains information required to Traverse between the Boxes when Navigating the Tree.
- The “Box-Link” Algorithms were found to be Much Faster than the “Suffix Tree” Approach. However, since the space required to store a Box-Link information is Much Higher than a Boolean array, the Box-Link approach was found to have a much higher space cost associated with storing the Box-Links for each node.

Example: A G T C G T C A T G A

A Suffix Tree is a Data Structure that represents all Suffixes of a string, allowing Fast Pattern Matching.

Steps:-

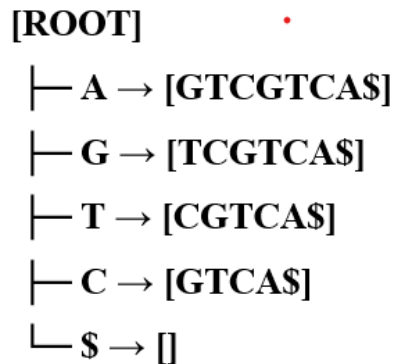
Generate All Suffixes

For the sequence AGTCGTCA, we generate all suffixes:

1. **AGTCGTCA**
2. **GTCGTCA**
3. **TCGTCA**
4. **CGTCA**
5. **GTCA**
6. **TCA**
7. **CA**
8. **A**

Construct the Tree

The Suffix Tree has one branch for each of the suffixes, and common prefixes are shared. Here's a simplified view:



The Suffix Tree allows for Fast Searches. For example, if we want to search for the Substring GT, the Tree would allow us to quickly navigate and find all positions of GT in the sequence.

Pattern 'GT' found at positions [2, 5].

Example: A G T C G T C A T G A

Box-Links are used in certain advanced algorithms for Efficient Pattern Matching. Here, we will consider them as a technique that divides the string into smaller “Boxes” to speed up Pattern searches.

Steps:-

Divide the Sequence into Boxes

For the sequence AGTCGTCA, we might divide it into segments (boxes),

[Box 1] AGT positions 1–3

Box 2 CGT positions 4–6

Box 3 TCA positions 6–8

Link Boxes

Each box has pointers or links to other boxes. Skip irrelevant boxes during pattern search.

[Box 1] × skip

Box 2 contains potential match (CGT)

Box 3 × skip

For Example, if we're searching for the substring GTC, we can quickly skip over Box 1 and Box 3 and focus only on Box 2.

Random and Uniform Projections

- Random Projections work by generating a projection of length k randomly, as described above. A hash table of size $4k$ is then created, which can be indexed by all possible strings of length k . Each entry of the hash table holds a bucket containing a number of strings. Every substring of length l from the set of input sequences is placed in the appropriate bucket in the hash table.
- Uniform Projection Method, improves this process even further, by increasing efficiency and reducing running times. This is achieved by considering a threshold for all projections, called as discrepancy value, which is the hamming distance between the maximum (over all projections) and each individual projection.
- “Uniform Projections” have been Found to be Better than “Random Projections” since the random algorithm works by considering the entire k -projection space. This tends to increase the running time, thus reducing efficiency. The Uniform algorithm samples a smaller space, based on the discrepancy value, and improves efficiency without affecting the accuracy.

Example: A G T C G T C A T G A

Steps:-

Map Sequence to Vector Space

First, represent your sequence as a vector. For example, encode the symbols A, G, T, C as integers or one-hot vectors:

A $\rightarrow$ [1, 0, 0, 0]

G $\rightarrow$ [0, 1, 0, 0]

T $\rightarrow$ [0, 0, 1, 0]

C $\rightarrow$ [0, 0, 0, 1]

Apply Random Projection

Use a random matrix to project the sequence into a lower-dimensional space.

For Random Projection:-

Gaussian Projection Matrix: We now apply a random projection to reduce the dimensions. This means multiplying the one-hot matrix with a projection matrix (Either Gaussian or Uniform). Values drawn from a normal (Gaussian) distribution (mean = 0, variance = 1):-

Formula:

Each element of the matrix is drawn from:

$$x \sim \mathcal{N}(0, 1)$$

Where:

- $\mathcal{N}(0, 1)$ = normal distribution with **mean = 0** and **standard deviation = 1**

A [0.5, -1.0]

G [-0.2, 1.2]

T [1.1, -0.4]

C [0.3, 0.7]

Apply Uniform Projection

Uniform Projection Matrix: Values are random from a Uniform distribution.

Formula:

Each element is drawn from:

$$x \sim \mathcal{U}(a, b)$$

Where:

- $\mathcal{U}(a, b)$ is a **uniform distribution**
- Typical values: $a = -1, b = 1$

A [0.7, -0.5]

G [-0.3, 0.8]

T [0.6, -0.2]

C [0.4, 0.9]

- **Integration with Deep Learning:** Leveraging deep neural networks (e.g., CNNs or Transformers) to identify complex, non-linear sequence motifs.
- **Single-Cell Data Applications:** Adapting motif discovery techniques to handle data from single-cell sequencing for more precise regulatory element identification.
- **Real-Time and Scalable Algorithms:** Developing faster, memory-efficient motif discovery algorithms to handle large-scale genomic datasets in real time.
- **Visualization Tools:** Enhancing motif visualization through interactive tools for better biological interpretation and user engagement.

Conclusion

- Motif discovery in biological sequences is a cornerstone of bioinformatics, with wide applications in understanding gene regulation, protein-DNA interactions, and evolutionary biology.
- The exploration of multiple methodologies—ranging from probabilistic models and Markov chains to suffix trees and box-link structures—highlights the diversity and depth of strategies available to researchers.
- While traditional methods provide robust foundations, emerging trends in machine learning and high-throughput biology promise to redefine motif discovery's accuracy, scalability, and biological relevance.

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Thank You!!